

INFORMATION SYSTEMS 3A

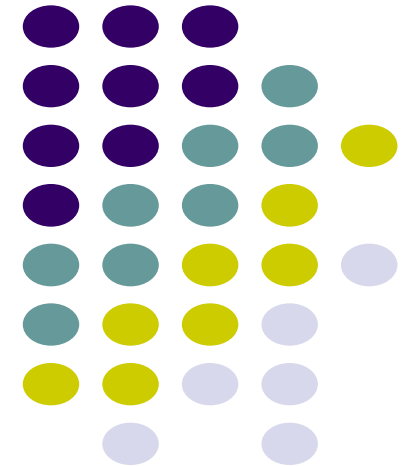
INSY7311

Learning Unit 6: Visual Design and Design issues

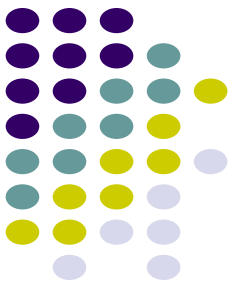


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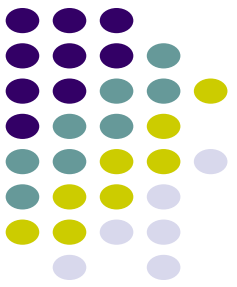
Amakan Elisha Agoni
eagoni@iie.ac.za



IIE Bachelor of Computer and Information Sciences in Application Development (BCA)



Learning Unit Objectives



Learning Unit 6: Visual Design and Design issues

Theme 1: Visual Design

LO1: Discuss Visual design.

LO2: Discuss visual elements.

LO3: Explain visual design principles.

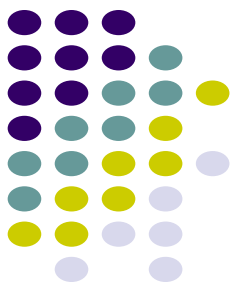
LO4: Explain gestalt principles and their use in design.

Theme 2: Visual Design languages & Styles

LO5: Explore the different design languages and styles.

LO6: Discuss issues that users can face due to design decisions.

Visual Design

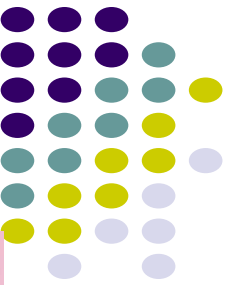


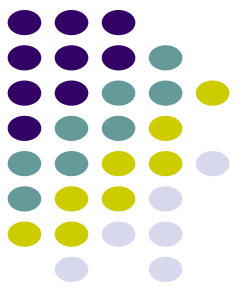
Introduction

- Visual design is a fundamental aspect of interaction design because it determines **how users perceive, interpret, and interact with digital systems**.
- While functionality determines what a system can do, visual design determines how easily users can understand and use that system.
- Visual design includes the use of: Colour, Typography, Layout, Images, Icons, Spacing, Visual hierarchy.
- These elements work together to create an interface that is **clear, attractive, usable, and meaningful**.
- In modern digital systems such as websites, desktop software, and mobile applications, visual design is not only about aesthetics but also about **usability, accessibility, and user experience (UX)**.
- This learning unit explores visual design concepts, principles, and issues that affect user interaction with digital interfaces.



Theme 1: Visual Design





Visual Design

Definition of Visual Design

- Visual design refers to the **appearance and presentation of information and interface elements within a digital system**. It focuses on how content is organized and displayed so that users can easily understand and interact with the system.

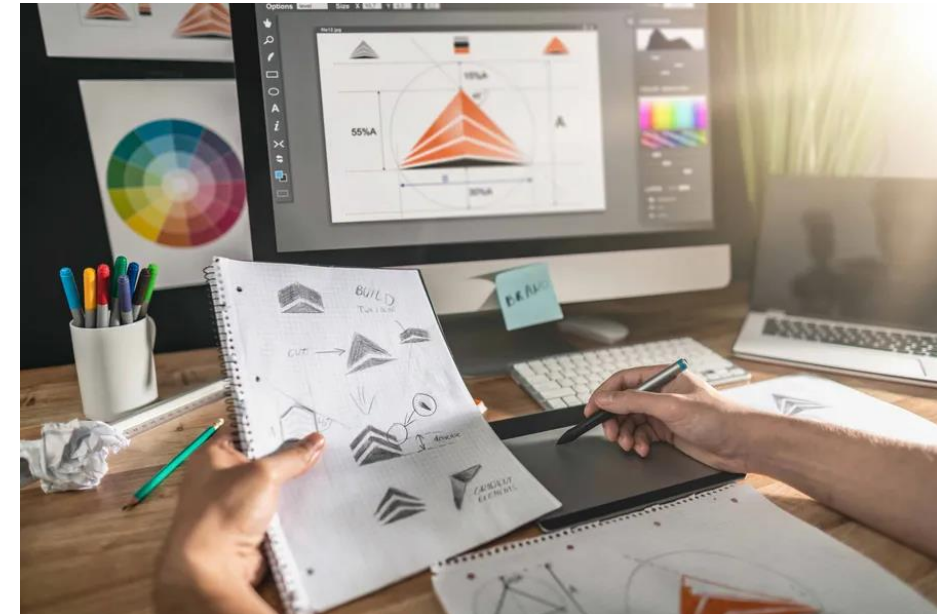
Visual design aims to achieve three primary goals:

- **Clarity** – ensuring information is easy to understand
- **Usability** – ensuring users can interact efficiently
- **Aesthetics** – ensuring the interface is visually appealing

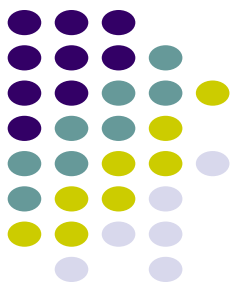
A well-designed interface allows users to quickly determine:

- What information is available
- Where they are in the system
- What actions they can take next

In contrast, poor visual design may cause confusion, misinterpretation, and errors.



Visual Design



Importance of Visual Design in Interaction Design

- Visual design influences several aspects of user interaction, including:

First Impressions

- Users often judge a system within seconds of viewing it. A visually appealing interface creates positive expectations and encourages continued interaction.

Readability and Comprehension

Visual design helps users quickly understand information through the use of:

- structured layouts
- headings
- icons
- visual hierarchy

Efficiency

- A good visual layout reduces the time users spend searching for information or functions.

Emotional Impact

- Design aesthetics can influence how users feel about a product.

Examples:

- Clean designs may feel professional and trustworthy.
- Colourful designs may feel fun and engaging.

Visual Elements

- Visual elements are the **building blocks of visual design**. They determine how the interface appears and how information is communicated.

Colour

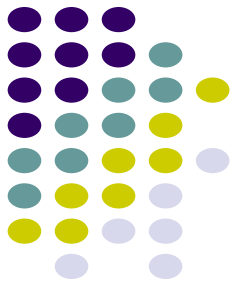
Colour plays a powerful role in interface design.

It can be used to:

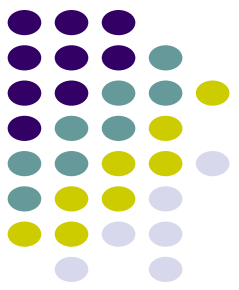
- highlight important elements
- indicate status or feedback
- group related items
- create emotional responses

Examples:

- **Green** → success
 - **Red** → error
 - **Blue** → links or actions
-
- Designers must also consider accessibility issues such as **colour blindness**, ensuring that important information is not communicated using colour alone.



Visual Elements



Typography

Typography refers to the **style and arrangement of text**.

Important typography factors include:

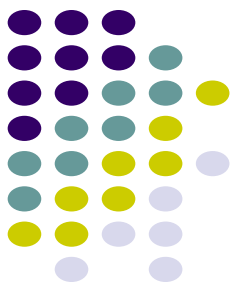
- font type
- font size
- Spacing
- line height
- readability

Good typography ensures that users can easily read and understand information.

Common design guidelines include:

- Use simple and readable fonts
- Maintain consistent font usage
- Ensure sufficient contrast between text and background





Visual Elements

Layout

Layout refers to **how interface elements are arranged on a screen.**

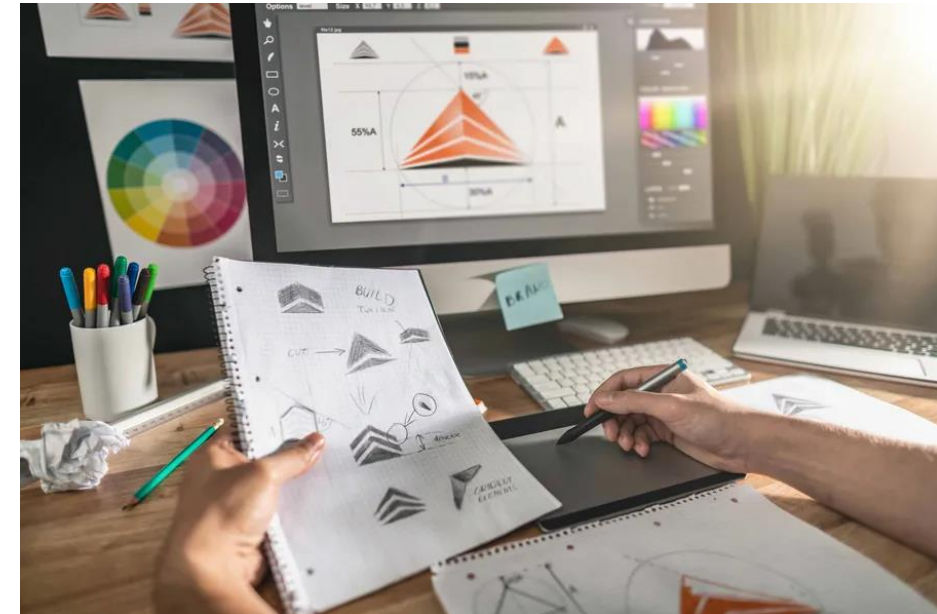
A well-structured layout helps users quickly identify:

- key content
- navigation options
- available actions

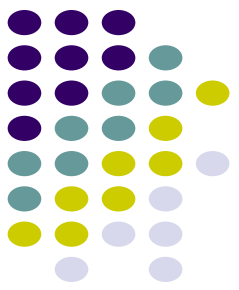
Good layouts follow a logical structure that aligns with users' expectations.

For example:

- navigation at the top
- main content in the centre
- additional information on the side



Visual Elements



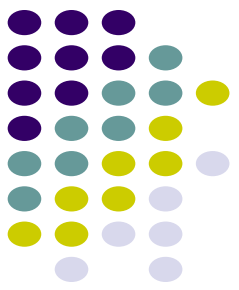
Icons

- Icons are graphical symbols representing actions, objects, or functions.

Examples include:

- search icon (magnifying glass)
- settings icon (gear)
- delete icon (trash bin)
- Icons help users recognize functions quickly without reading text.
- However, icons must be designed carefully so that their meaning is clear and recognizable.





Visual Elements

Images and Graphics

- Images can enhance communication by illustrating concepts visually.

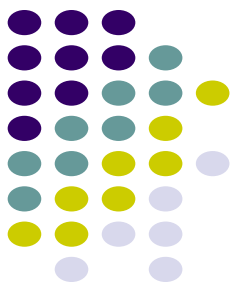
They may include:

- Photographs
- Diagrams
- Illustrations
- Infographics

Images should be used meaningfully and not simply for decoration.



Visual Elements



Spacing (Whitespace)

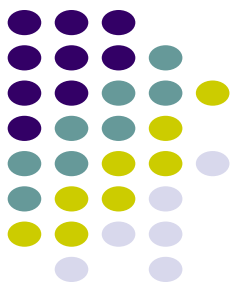
- Whitespace refers to the empty space around elements in an interface.

It improves readability by:

- separating different sections
- reducing clutter
- improving focus on important content

Whitespace is a critical component of modern minimalist design.





Visual Elements

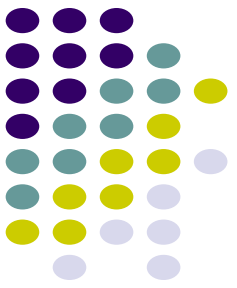
Visual Hierarchy

- Visual hierarchy refers to **arranging elements so that users notice the most important information first.**
- This can be achieved using:
 - larger font sizes
 - bold text
 - colour contrast
 - Positioning
 - Spacing

For example:

- Headings are larger than body text to indicate importance.





Visual Design Principles

- Visual design principles guide how visual elements should be arranged and used effectively.

Balance

- Balance refers to the distribution of visual weight across the interface.
- Two common types include:

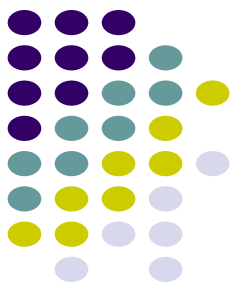
Symmetrical balance

- Both sides of the screen appear evenly distributed.

Asymmetrical balance

- Elements are arranged unevenly but still create visual stability.





Visual Design Principles

Alignment

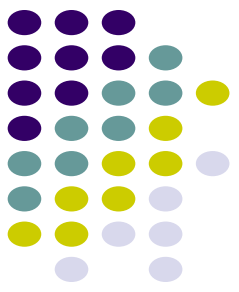
- Alignment ensures that interface elements line up consistently.

Proper alignment:

- improves readability
- creates visual order
- reduces clutter

Misaligned elements can make an interface appear messy and confusing.





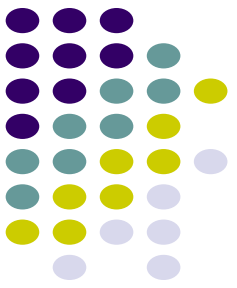
Visual Design Principles

Consistency

- Consistency ensures that design elements remain uniform across the interface.
- Examples include consistent:
 - Colours
 - Fonts
 - button styles
 - navigation placement

Consistency allows users to predict how the system behaves.





Visual Design Principles

Contrast

- Contrast helps differentiate elements and emphasize important information.

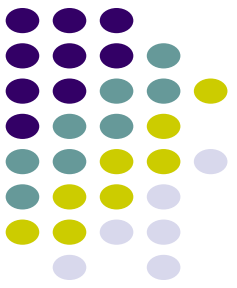
Examples include contrast in:

- Colour
- Size
- Typography
- brightness

For example:

- A dark button on a light background attracts attention.





Visual Design Principles

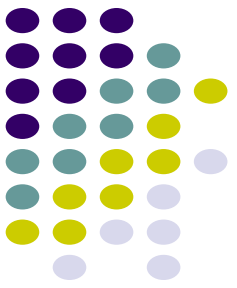
Proximity

- Proximity refers to placing related elements close together.
- This helps users understand relationships between different pieces of information.

Example:

- Form labels should appear near the input fields they describe.





Visual Design Principles

Simplicity

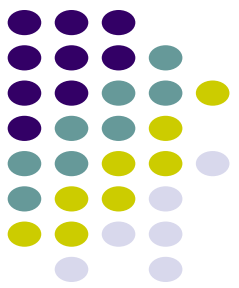
Interfaces should avoid unnecessary complexity.

Simple designs:

- reduce cognitive load
- improve usability
- make navigation easier

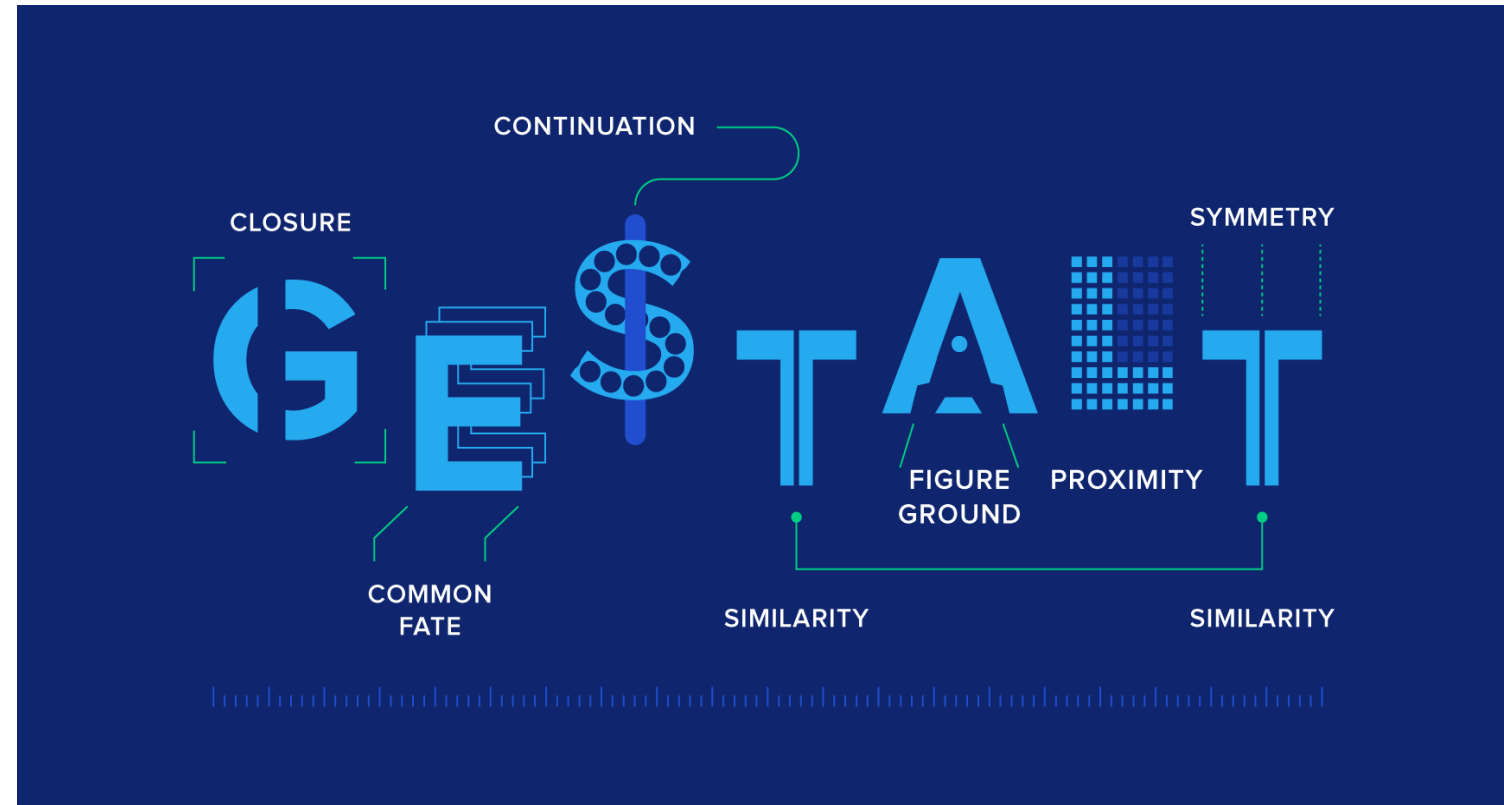
This principle is often referred to as minimalist design.





Gestalt Principles and Their Use in Design

- Gestalt psychology explains how humans perceive visual information as organized patterns rather than separate elements.
 - These principles help designers create intuitive and easily understandable interfaces.
1. Law of Proximity
 2. Law of Similarity
 3. Law of Continuity
 4. Law of Closure
 5. Figure–Ground Relationship



Gestalt Principles and Their Use in Design

Law of Proximity

- Elements that are close together are perceived as related.

Example:

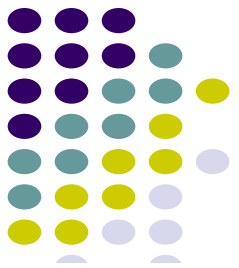
- Grouping menu items together indicates that they belong to the same category.

Law of Similarity

- Elements that look similar are perceived as belonging together.

Example:

- Buttons with the same colour and shape indicate similar functions.



PROXIMITY

When objects placed together, the eye perceives them as a group.



SIMILARITY

When objects look similar to one another, the eye perceives them as a group or pattern.



CONTINUANCE

The eye is compelled to move from one object through another.



CLOSURE

When an object is incomplete or not completely enclosed.



FIGURE & GROUND

When the eye differentiates an object from its surrounding area.

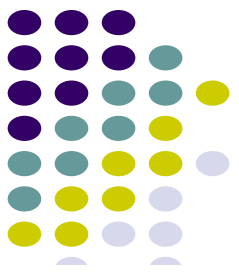
Gestalt Principles and Their Use in Design

Law of Continuity

- People naturally follow lines or curves when viewing visual elements.
- Designers can guide user attention by arranging elements in a continuous flow.
- **Example:**
- Step-by-step processes arranged vertically.

Law of Closure

- People tend to fill in missing parts of incomplete shapes.
- **Example:**
- Logos often use partial shapes that the brain automatically completes.



PROXIMITY

When objects placed together, the eye perceives them as a group.



SIMILARITY

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CONTINUANCE

The eye is compelled to move from one object through another.



CLOSURE

When an object is incomplete or not completely enclosed.



FIGURE & GROUND

When the eye differentiates an object from its surrounding area.

Gestalt Principles and Their Use in Design

Figure–Ground Relationship

- Users distinguish objects (figure) from the background.
- Good design ensures that important elements stand out clearly.

Example:

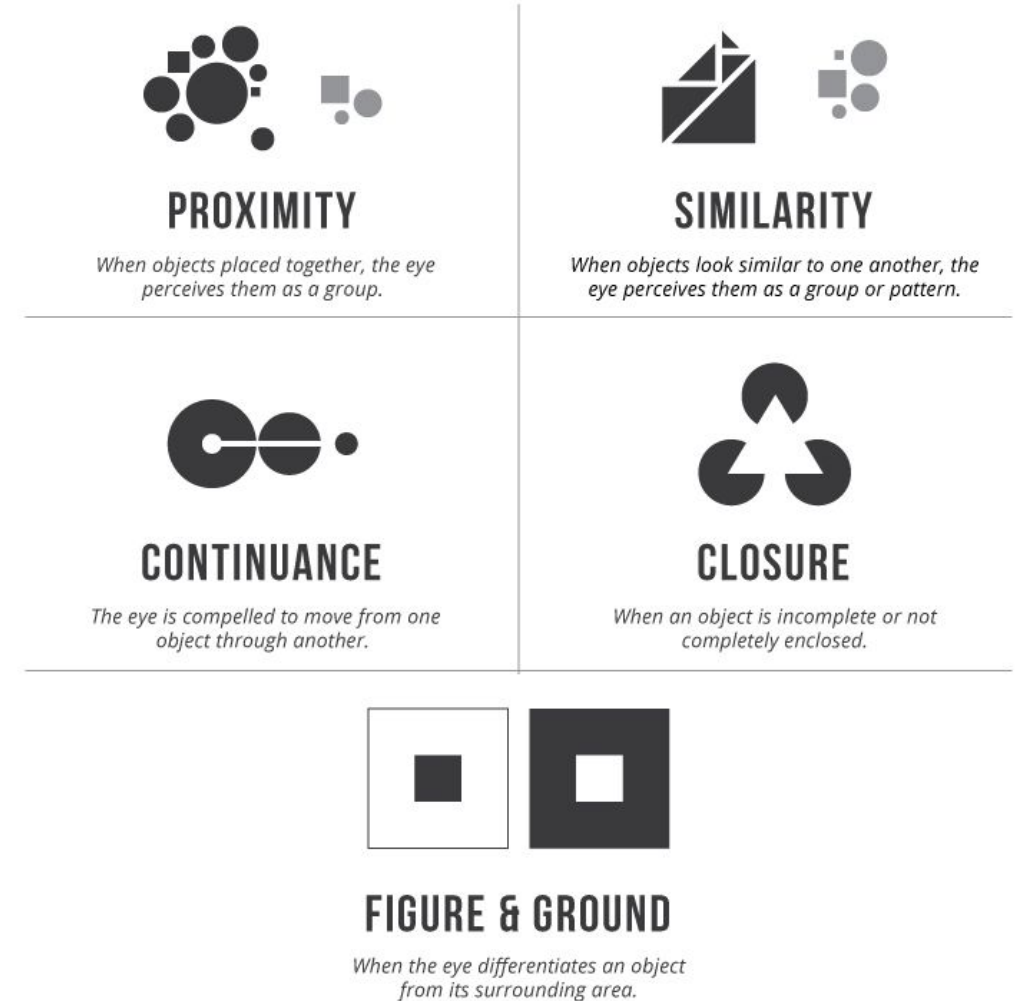
- Dark text on a light background.

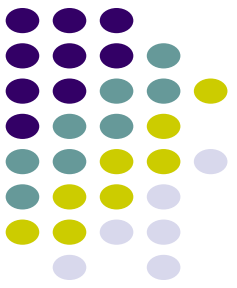
Application of Gestalt Principles in Interface Design

Gestalt principles help designers:

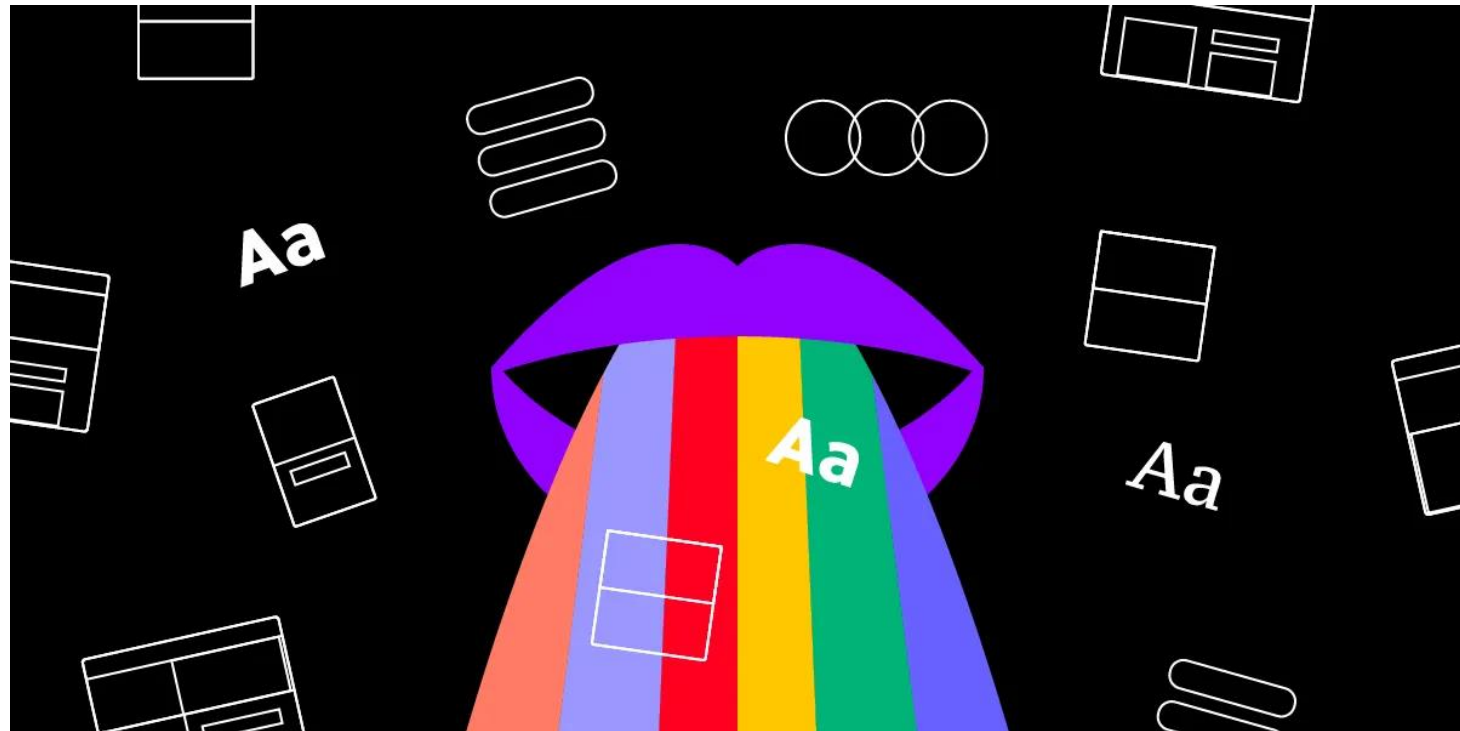
- organize information clearly
- create intuitive layouts
- improve visual communication

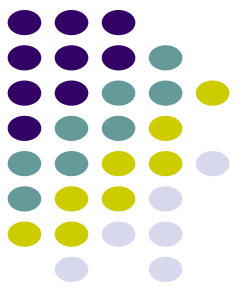
They are widely used in website layouts, mobile interfaces, and dashboards.





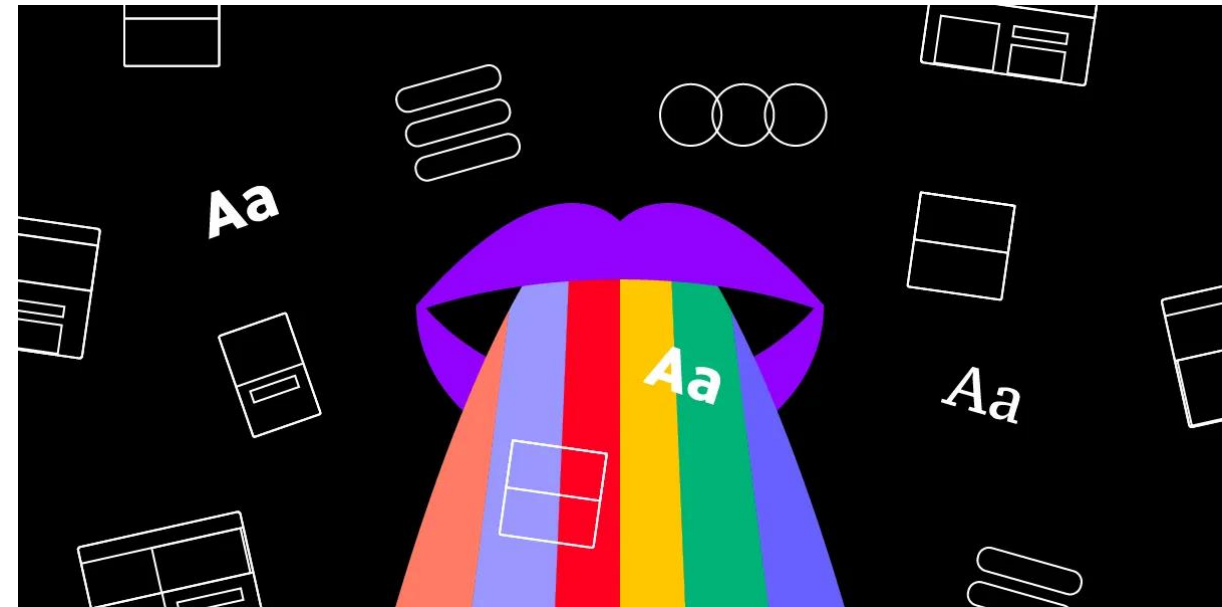
Theme 2: Visual Design Languages and Styles

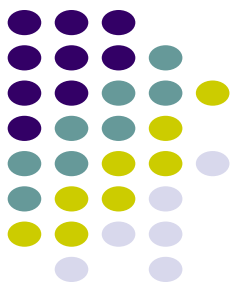




The Different Design Languages and Styles

- A design language is a **consistent set of visual design rules and components used across products or platforms.**
 - Design languages ensure that interfaces remain consistent and recognizable.
1. Skeuomorphic Design
 2. Flat Design
 3. Material Design
 4. Responsive Design
 5. Minimalist Design





The Different Design Languages and Styles

Skeuomorphic Design

- Skeuomorphic design mimics real-world objects.

Examples include:

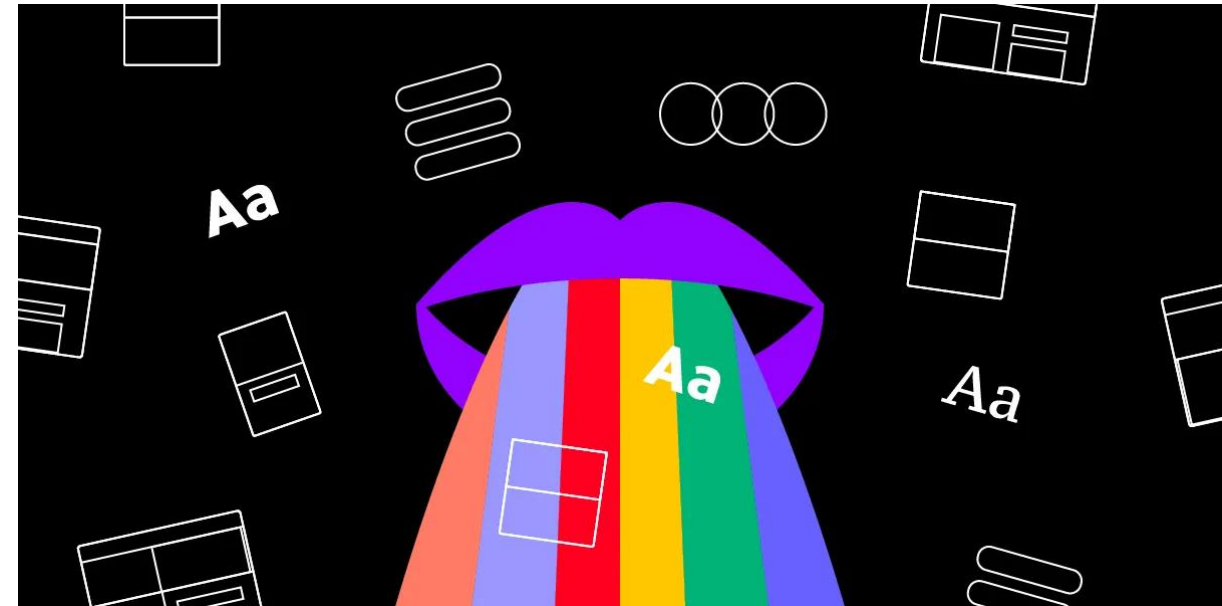
- digital calendars that resemble paper calendars
- calculator apps that look like physical calculators

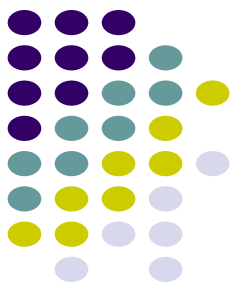
Advantages:

- familiar to users
- easy for beginners to understand

Disadvantages:

- may waste screen space
- can appear outdated





The Different Design Languages and Styles

Flat Design

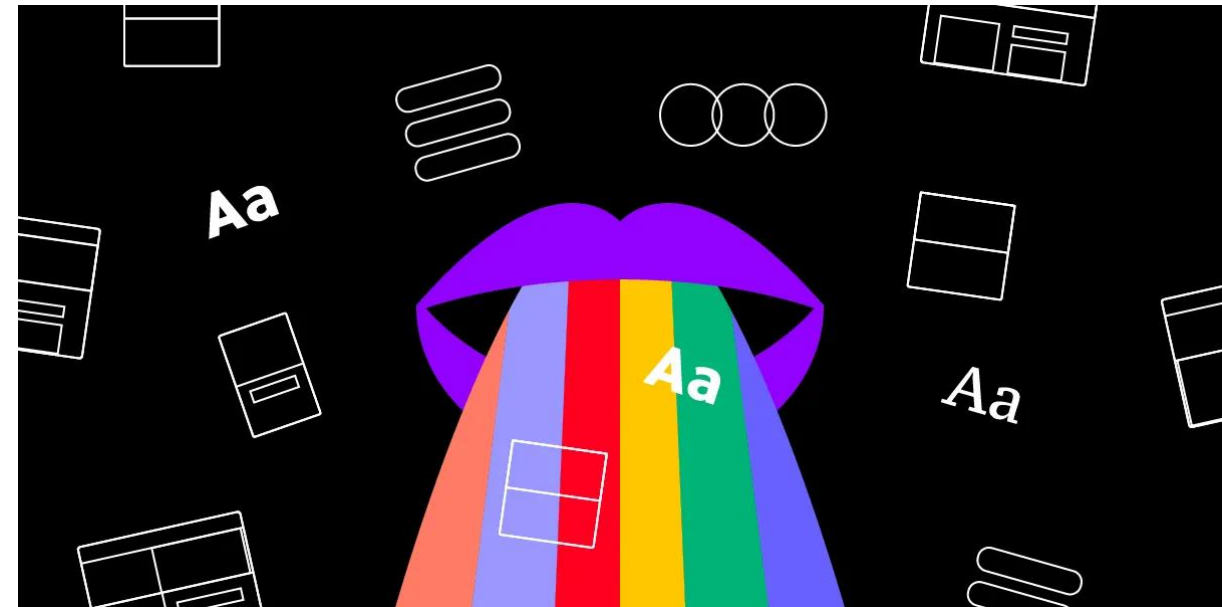
- Flat design removes unnecessary decorative elements such as shadows or textures.

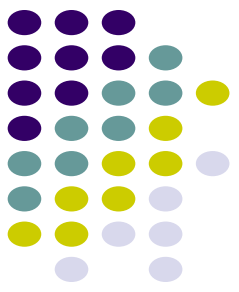
Characteristics include:

- simple shapes
- bright colours
- minimal visual effects

Advantages:

- clean appearance
- faster loading times
- easier responsiveness





The Different Design Languages and Styles

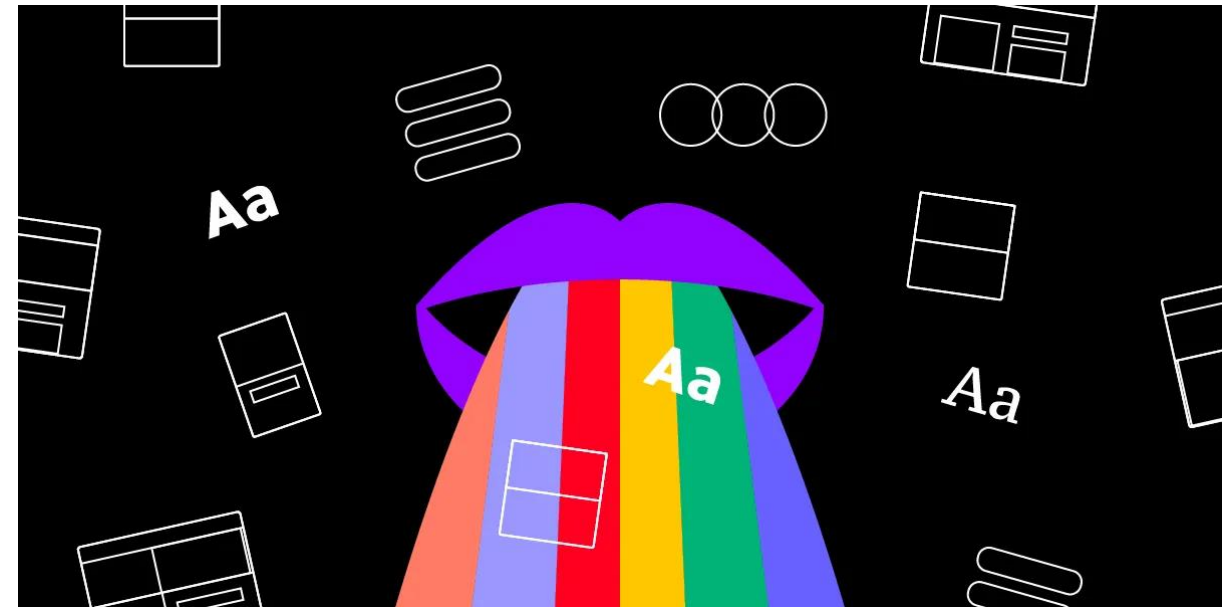
Material Design

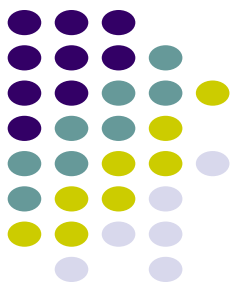
- Material design is a design language developed by Google.
- It combines flat design with subtle depth effects such as shadows and layering.

Features include:

- responsive layouts
- animated transitions
- consistent components

Material design helps create visually appealing and functional interfaces.





The Different Design Languages and Styles

Responsive Design

- Responsive design ensures that interfaces adapt to different screen sizes.

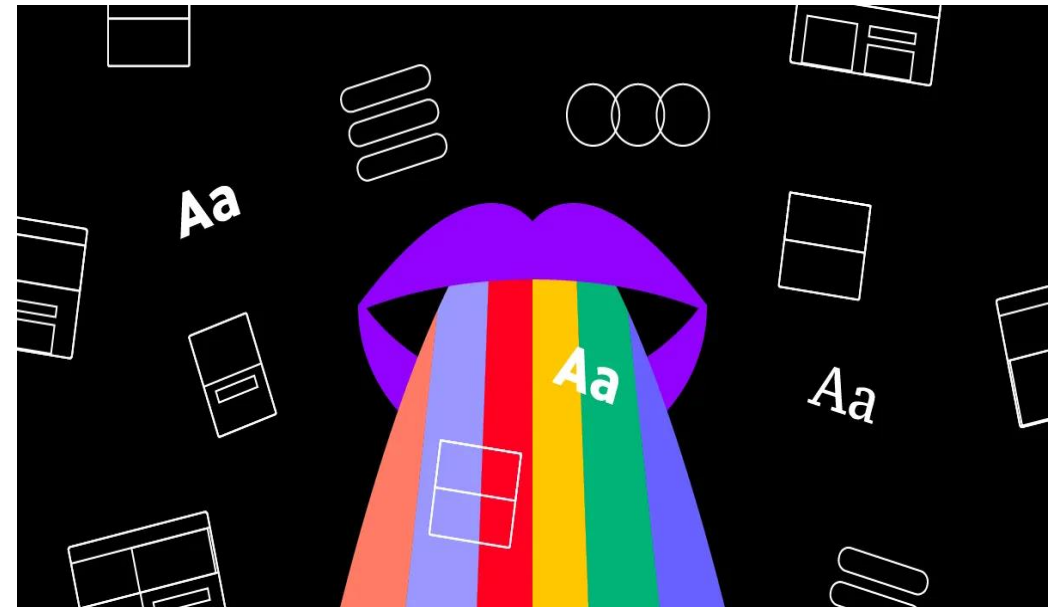
Examples include:

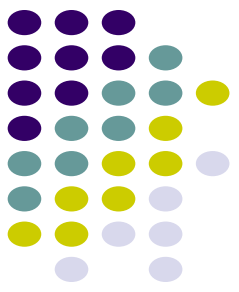
- desktop screens
- Tablets
- Smartphones

Responsive layouts automatically adjust elements such as:

- Navigation
- font size
- images

This allows a single system to function effectively across multiple devices.





The Different Design Languages and Styles

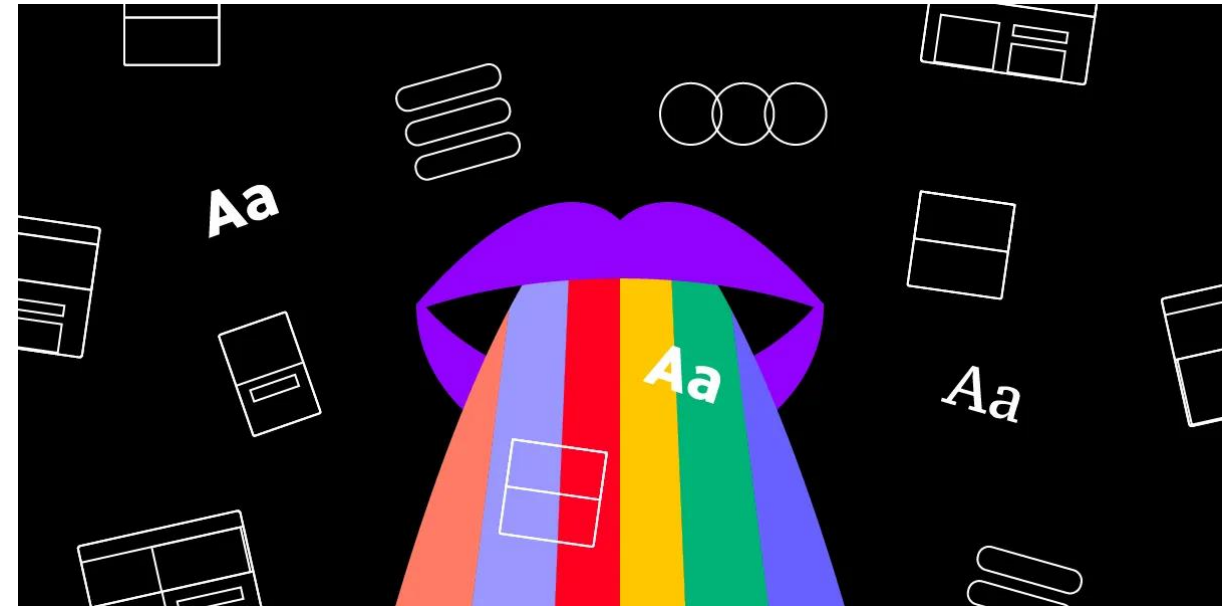
Minimalist Design

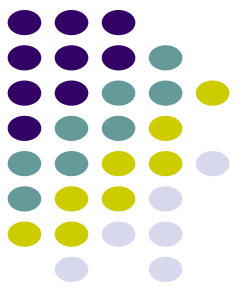
- Minimalist design focuses on simplicity.

Characteristics include:

- limited colour palette
- simple typography
- large whitespace areas

Minimalist interfaces reduce distractions and improve usability.





Issues that Users Can Face Due to Design Decisions

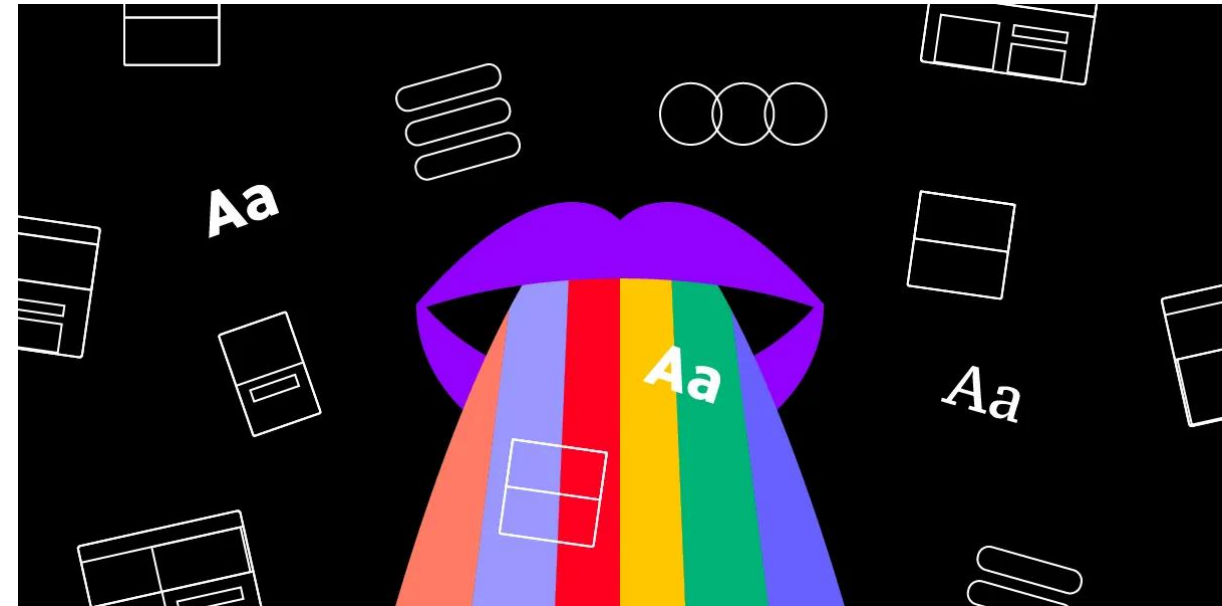
- Poor design decisions can negatively affect user experience.
- Some common problems include:

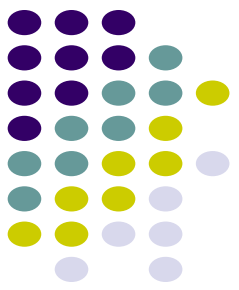
Visual Clutter

- Too many elements on a screen can overwhelm users.

Effects include:

- difficulty locating information
- reduced readability
- increased cognitive load

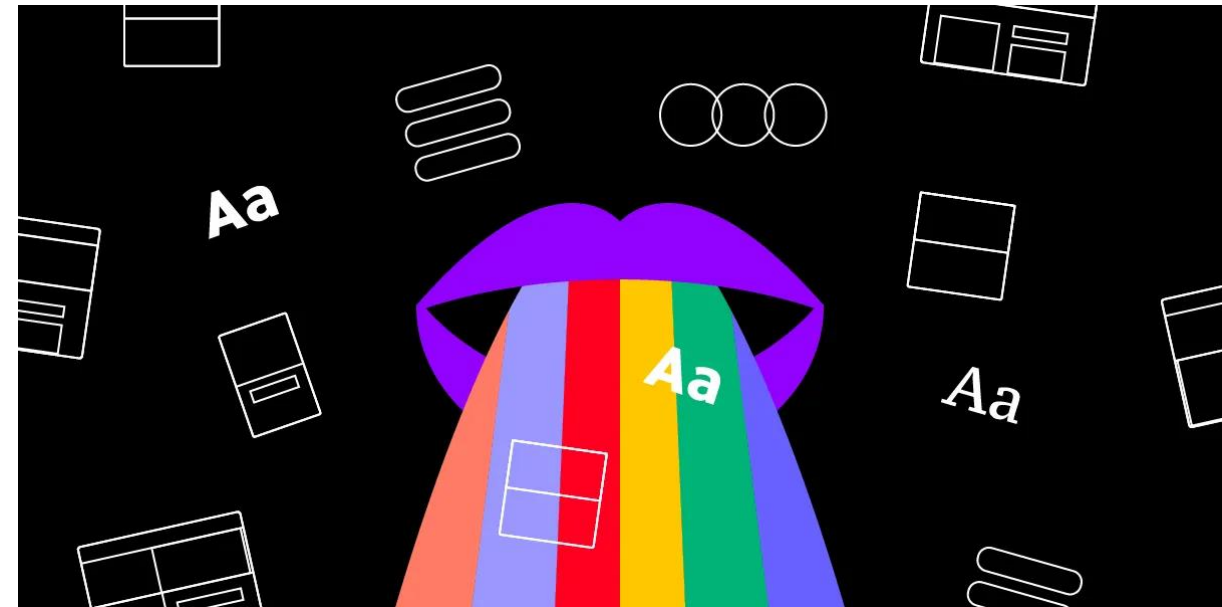


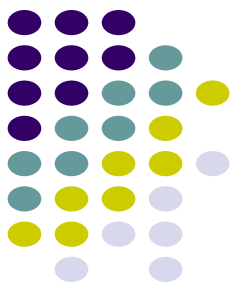


Issues that Users Can Face Due to Design Decisions

Poor Colour Choices

- Incorrect colour combinations can cause problems such as:
- poor contrast
- readability issues
- accessibility problems
- Designers must consider users with visual impairments.





Issues that Users Can Face Due to Design Decisions

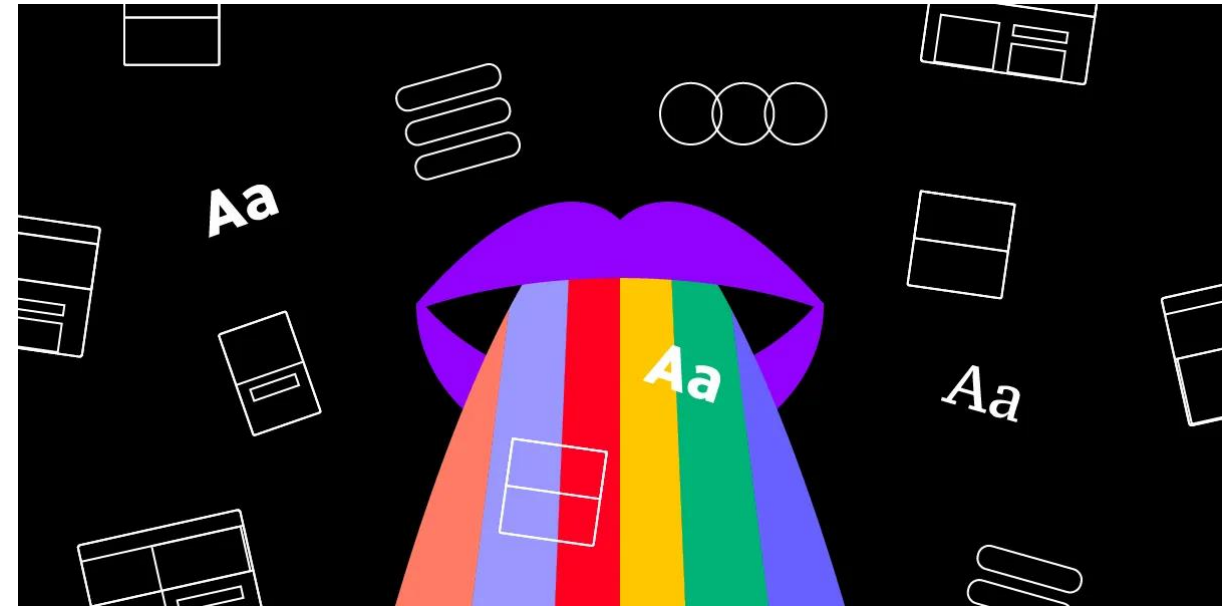
Inconsistent Design

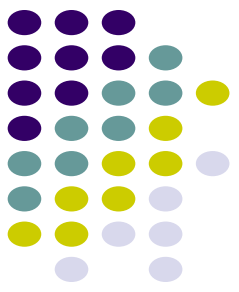
- When elements change across screens, users become confused.

Examples:

- buttons changing colour
- navigation appearing in different places

Consistency is essential for usability.

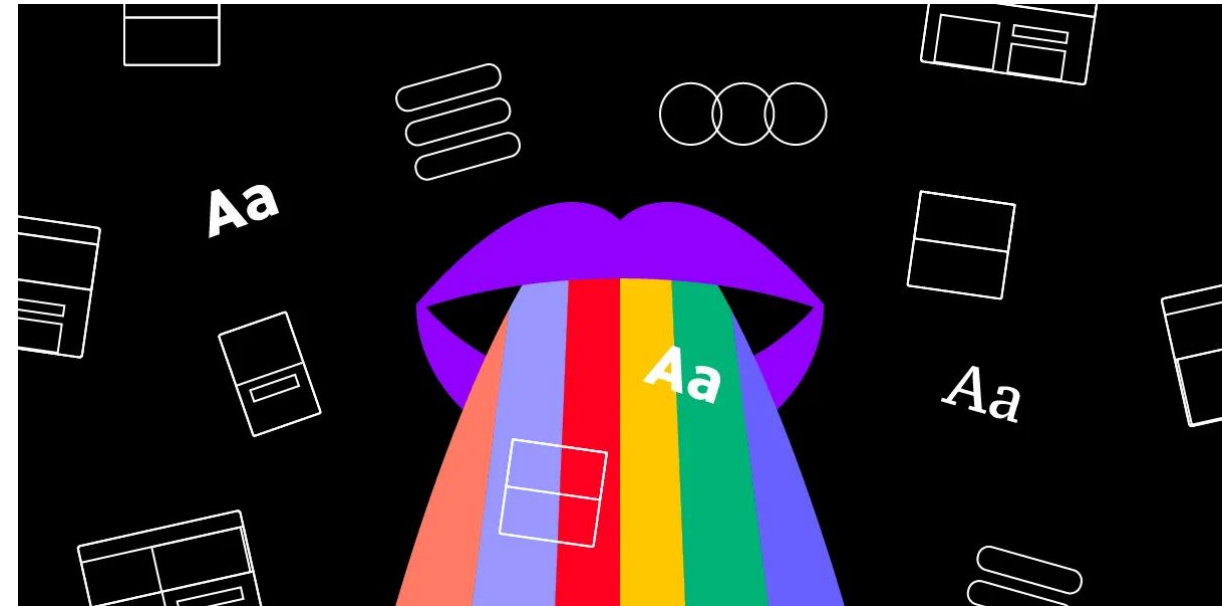


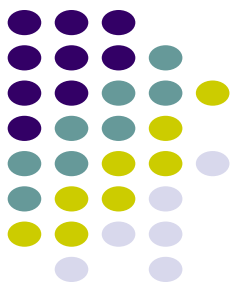


Issues that Users Can Face Due to Design Decisions

Poor Typography

- Text that is too small, crowded, or decorative can reduce readability.
- Users may struggle to read important information.

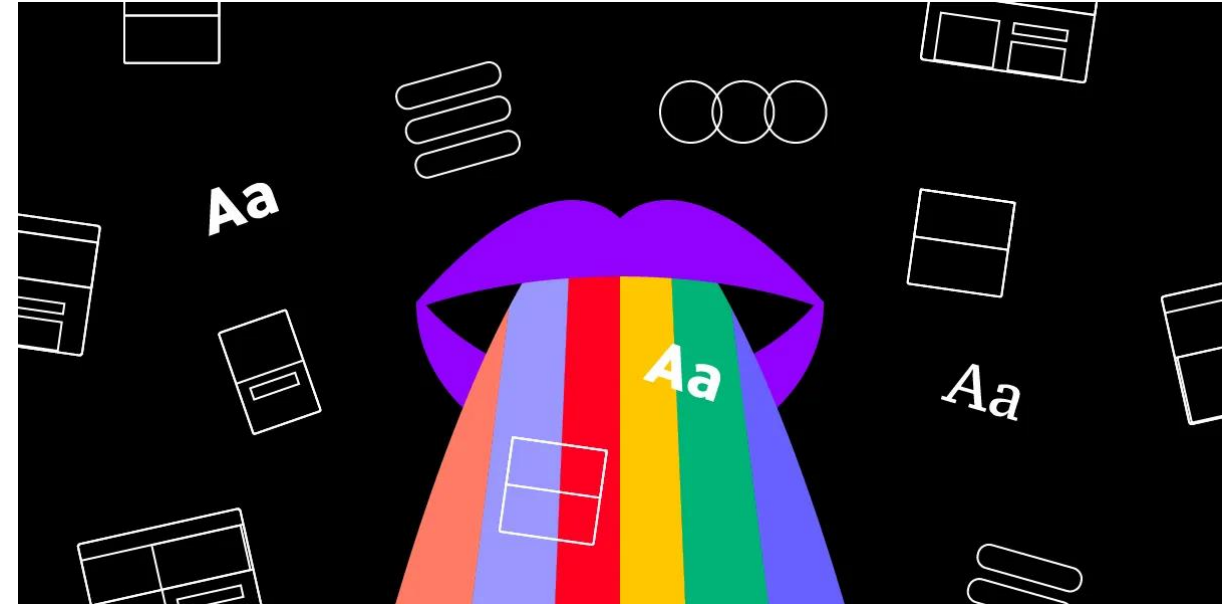


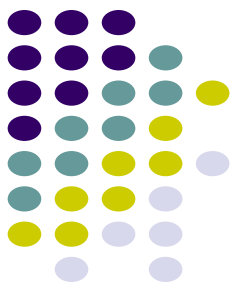


Issues that Users Can Face Due to Design Decisions

Overuse of Animation

- Animations can improve engagement, but excessive animations can:
- distract users
- slow down interaction
- cause usability problems





Issues that Users Can Face Due to Design Decisions

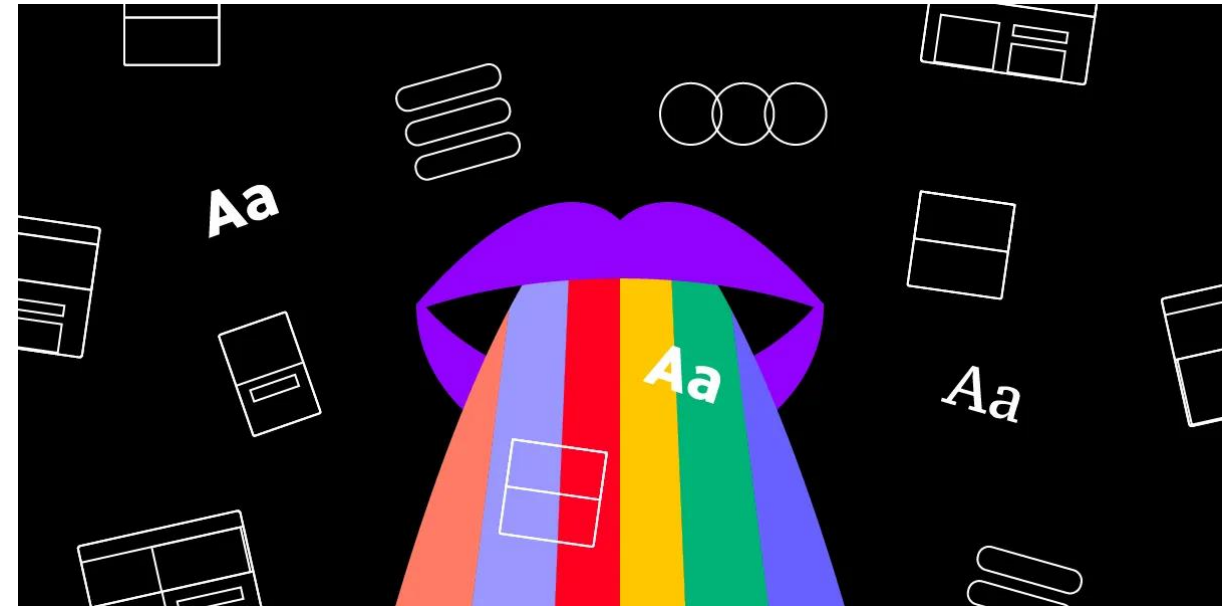
Accessibility Issues

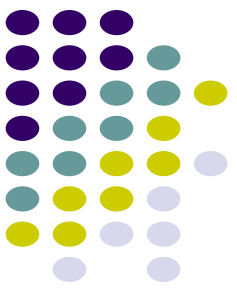
- Design decisions may unintentionally exclude certain users.

Examples include:

- small text for visually impaired users
- colour-only indicators
- poor contrast ratios

Designers must ensure systems are usable by diverse users.





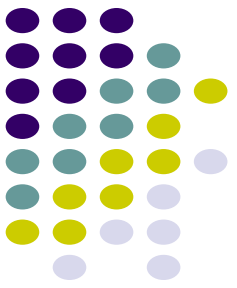
Issues that Users Can Face Due to Design Decisions

Conclusion

- Visual design is an essential component of interaction design because it shapes how users perceive and interact with digital systems.

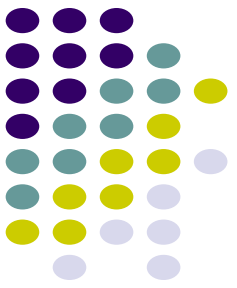
Effective visual design combines:

- clear visual elements
 - sound design principles
 - psychological insights such as Gestalt principles
 - consistent design languages
-
- When applied correctly, visual design improves usability, efficiency, and user satisfaction.
 - However, poor design decisions can create barriers that make systems confusing, frustrating, or inaccessible.
 - For this reason, designers must carefully consider visual design principles to ensure that interfaces are **clear, functional, and user-friendly.**



Next Class:

Learning Unit 7: Evaluation and Usability Testing



Thank you for listening



Any question?